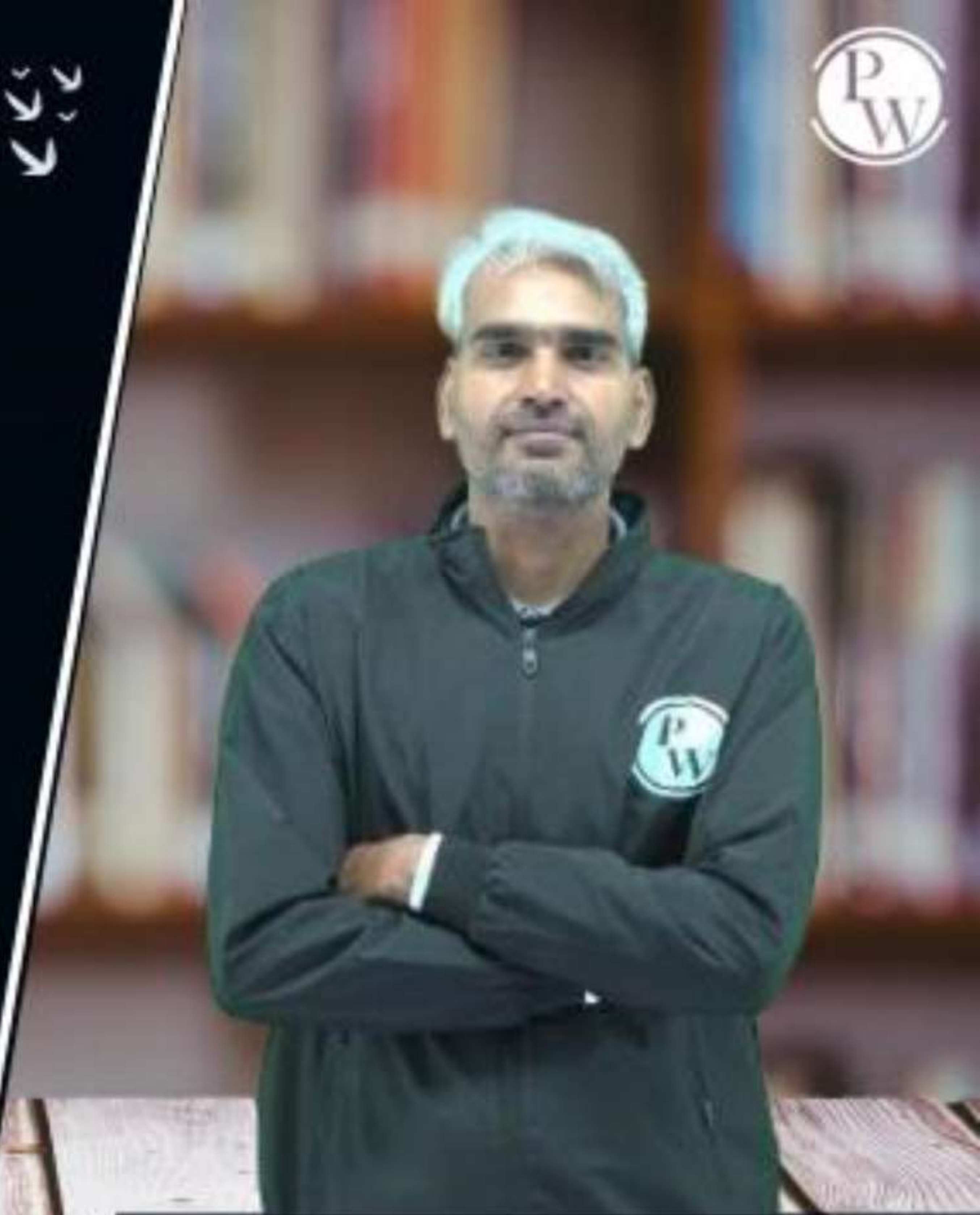
# Computer Science & IT

## Discrete Mathematics

**Set Theory & Algebra**

**Lecture No. 05**



By- Vishal Sir

# Recap of Previous Lecture

- Topic: Cartesian product — "$A \times B$"
- Topic: Relation — # Relations from A to B $= 2^{|A \times B|}$
- Topic: Different types of relations
- Topic: Diagonal relation
- Topic: Reflexive relation

Slide

# Topics to be Covered

- **Topic** Irreflexive relation
- **Topic** Symmetric relation
- **Topic** Anti-symmetric relation
- **Topic** Asymmetric relation

Slide

# Topic : Irreflexive Relation

A relation R on set A is called irreflexive relation if and only if, $a \not R a, \forall a \in A$

i.e. $(a, a) \notin R, \forall a \in A$

i.e., no diagonal order pair should be present in the relation

Slide

eg: Let, $A = \{1, 2, 3\}$

① $\underbrace{R_1 = \{\ \ \}}_{\text{Empty relation}}$ Empty relation is the smallest irreflexive relation on any set A.

② $R_2 = \{(1,2), (2,3), (3,1)\}$ it is an irreflexive relation

③ $R_3 = \{(1,2), (2,1), (2,2), (3,2)\}$ $R_3$ is not reflexive as $(1,1), (3,3) \notin R_3$

$R_3$ is not irreflexive because $(2,2) \in R_3$

$R_3$ is neither reflexive nor irreflexive

Q:- Let $A = \{1, 2, 3, \ldots\ldots, n\}$

How many "irreflexive relation" are possible on set A

where $|A| = n$.

Q: Let $A = \{1, 2, 3, \ldots, n\}$

How many "irreflexive relation" are possible on set A where $|A| = n$.

Irreflexive Relation = (None of the diagonal order pair should be present) and (any number of non-diagonal order pairs may be present)

Choose all '0' out of 'n' diagonal order pairs

Choose any number of non-diagonal order pairs out of $(n^2-n)$

$\therefore$ Number of irreflexive $Rel^n$ =

$$\left({}^{n}C_{0}\right) * \left\{ {}^{n^2-n}C_0 + {}^{n^2-n}C_1 + {}^{n^2-n}C_2 + \cdots + {}^{(n^2-n)}C_{(n^2-n)} \right\}$$

$$= 1 * 2^{n^2-n}$$

$$\boxed{= 2^{n^2-n}}$$

# Topic : Symmetric Relation

A relation 'R' on set A is called a symmetric relation if and only if,

$\left(\text{if } a \, R \, b \text{ then } b \, R \, a\right), \forall a, b \in A$

i.e. $\left(\text{if } (a, b) \in R \text{ then } (b, a) \in R\right), \forall a, b \in A$

Slide

eg: Let, $A = \{1, 2, 3\}$

① $R_1 = \{\ \}$ Empty relation is the smallest symmetric relation on any set A.

② $R_2 = \{(1,1)\}$ it is symmetric { Because of Presence of absence of any of the diagonal order Pair, there will be no problem w.r.t. Symmetricity Property }

③ $R_3 = \{(1,2), (2,1), (2,2)\}$ } $R_3$ is symmetric

diagonal order pair ∴ No Problem

both $(a,b)$ & $(b,a) \in R_3$ ∴ No problem

④ $R_4 = \{(1,1), (2,3), (3,1), (3,2)\}$

No problem

$(3,1) \in R_4$ but $(1,3) \notin R_4$ ∴ $R_4$ is not Symmetric

**Note:**

1. Empty relation is the smallest symmetric relation

2. "$A \times A$" is the largest symmetric $rel^n$ on set $A$.

Q:- Let $A = \{1, 2, 3, \ldots\ldots, n\}$

How many "symmetric relation" are possible on set A

where $|A| = n$.

Let $A = \{1, 2, 3, 4, \ldots\ldots, n\}$

Pair of symmetric non-diagonal order pairs

$$A \times A = \left\{ \begin{matrix} (1,1), & (1,2), & (1,3), & \ldots\ldots & (1,n) \\ (2,1), & (2,2), & (2,3), & \ldots\ldots & (2,n) \\ (3,1), & (3,2), & (3,3), & \ldots\ldots & (3,n) \\ \vdots & \vdots & \vdots & & \vdots \\ (n,1), & (n,2), & (n,3), & \ldots\ldots & (n,n) \end{matrix} \right\}$$

no problem because of Presence of absence of diagonal order pair.

- In a symmetric relation, either both the order Pairs of "Pair of symmetric non-diagonal order pairs" must be present, or none of them should be present.

# Pairs of symmetric non-diagonal order pairs $= \dfrac{(n^2-n)}{2}$

Q:- Let $A = \{1, 2, 3, \dots, n\}$

How many "symmetric relation" are possible on set A where $|A| = n$.

Symmetric Relation = (Any number of diagonal orde pairs may be present) and (any number of pairs of symmetric non-diagonal order pair may be present)

$\therefore$ Number of Symmetric Relation $= 2^n * 2^{\left(\frac{n^2-n}{2}\right)}$

$= 2^{n+\frac{(n^2-n)}{2}} = 2^{\left(\frac{n^2+n}{2}\right)} = 2^{\frac{n(n+1)}{2}}$ Ans

## Another approach w.r.t. non-diagonal order pairs

- In a non-diagonal order pair $(a, b)$, we know that $a \neq b$

  if $a = b$, then $(a, b)$ is a diagonal order pair.

- For a non-diagonal order pair we will require two distinct elements 'a' and 'b' s.t. $a \neq b$

- W.r.t. two distinct elements 'a' and 'b' possible combinations of order pairs in an arbitrary relation

| Case (i) | Case (ii) | Case (iii) | Case (iv) |
| --- | --- | --- | --- |
| both $(a,b)$, $(b,a)$ are present | $(a,b)$ is present but $(b,a)$ is not present | $(a,b)$ is not present but $(b,a)$ is present | Both $(a,b)$ and $(b,a)$ are not present |

$\underbrace{\hspace{10em}}$

These are the only possible order pairs in a relation w.r.t. two distinct elements

## Another approach w.r.t. non-diagonal order pairs

- In a non-diagonal order pair $(a,b)$, we know that $a \neq b$

  if $a=b$, then $(a,b)$ is a diagonal order pair.

- For a non-diagonal order pair we will require two distinct elements 'a' and 'b' s.t. $a \neq b$

- W.r.t. two distinct elements 'a' and 'b' possible combinations of order pairs in an arbitrary relation

| Case (i) | Case (ii) | Case (iii) | Case (iv) |
|---|---|---|---|
| both $(a,b)$, $(b,a)$ are present | $(a,b)$ is present but $(b,a)$ is not present | $(a,b)$ is not present but $(b,a)$ is present | Both $(a,b)$ and $(b,a)$ are not present |
| allowed in Symmetric Rel$^n$ | Not allowed in a Symmetric Relation | Not allowed in a Symmetric Relation | Allowed in Symmetric Rel$^n$ |

Q:. Let $A = \{1, 2, 3, \ldots, n\}$

How many "symmetric relation" are possible on set A where $|A| = n$.

Number of Symmetric Rel$^n$ = (Possible Combinations w.r.t. diagonal order pair) and (Possible Combinations w.r.t. non-diagonal order pair)

$$= 2^n * 2^{\left({}^nC_2\right)}$$

Out of 'n' distinct elements, two distinct elements can be chosen in ${}^nC_2$ ways

Possible Choices of order pairs in a Symmetric Relation Corresponding to each pair of distinct elements

$$= 2^n * 2^{\frac{n(n-1)}{2}}$$

$$= 2^{\frac{n(n+1)}{2}}$$ Ans

$\{a, b, c, d\}$

a & b    a & c    a & d    b & c    b & d    c & d

$\{(a,b), (b,a)\}$ $\times$ 2 $\times$ 2 $\times$ 2 $\times$ 2 $\times$ 2

(or)

$\{\cancel{(a,b)}, \cancel{(b,a)}\}$

Two Choices of order Pairs in a Symmetric Relⁿ

# Topic : Anti-Symmetric Relation

A relation R on set A is called an anti-symmetric relation if and only if,

$\left(\text{if } a\,R\,b \text{ and } b\,R\,a, \text{ then } a=b\right), \forall a,b \in A$

i.e. $\left(\text{if } (a,b) \in R \text{ and } (b,a) \in R, \text{ then } a=b\right), \forall a,b \in A$

if $a=b$, then $(a,b)$ is diagonal order pair.

$(a,b) \in R$ and $(b,a) \in R$, it should happen only w.r.t. diagonal order pairs

otherwise, if $a \neq b$, then if $(a,b) \in R$ then $(b,a) \notin R$

Slide

Note:- In an anti-symmetric relation there will be no problem because of presence or absence of diagonal order Pairs

if $a \neq b$, i.e. when "then" Condition is false for "if - then" statement to be true the "if" condn itself must be false.

i.e. if $a \neq b$ then (i) Either $(a,b) \notin R$

(ii) or $(b,a) \notin R$

(iii) or both $(a,b) \notin R$ and $(b,a) \notin R$

eg: Let $A = \{1, 2, 3\}$

① $R_1 = \{\ \}$ it is the smallest anti-symmetric relation on set A.

② $R_2 = \{(1,1), (3,3)\}$ it is an anti-symmetric relation

③ $R_3 = \{(1,1), (1,2), (3,2)\}$ it is an anti-symmetric $Rel^n$

diagonal

$(3,2) \in R_3$ but $(2,3) \notin R_3$ ∴ if statement is false and hence "if-then" is true

$(1,2) \in R_3$ but $(2,1) \notin R_3$ ∴ if statement is false and hence "if-then" is true

④ $R_4 = \{(1,2), (2,3), (3,1), (3,2)\}$

$\underbrace{(2,3) \in R_4 \ \& \ (3,2) \in R_4}_{\text{if true}}$ but $\underbrace{2 \neq 3}_{\text{then "false"}}$

∴ $R_4$ is not an Anti-symmetric $Rel^n$.

**Note:** If a relation does not contain any non-diagonal order pair, then every such relation is symmetric as well as anti-symmetric

Q: Let $A = \{1, 2, 3, \ldots, n\}$

How many "anti-symmetric relation" are possible on set A where $|A| = n$.

if $(a \neq b)$ then possible order pairs of a & b that are allowed in an anti-symmetric relation

$(a,b) \in R$ but $(b,a) \notin R$

$(a,b) \notin R$ and $(b,a) \in R$

$(a,b) \notin R$ and $(b,a) \notin R$

3 Combinations are allowed w.r.t two distinct elements

$(a,b) \in R$ and $(b,a) \in R$ not allowed

## Another approach w.r.t. non-diagonal order pairs

- In a non-diagonal order pair $(a,b)$, we know that $a \neq b$

  if $a=b$, then $(a,b)$ is a diagonal order pair.

- For a non-diagonal order pair we will require two distinct elements 'a' and 'b' s.t. $a \neq b$

- W.r.t. two distinct elements 'a' and 'b' possible combinations of order pairs in an arbitrary relation

| Case (i) | Case (ii) | Case (iii) | Case (iv) |
| --- | --- | --- | --- |
| both $(a,b)$, $(b,a)$ are present | $(a,b)$ is present but $(b,a)$ is not present | $(a,b)$ is not present but $(b,a)$ is present | Both $(a,b)$ and $(b,a)$ are not present |
| Not allowed in an Anti-Symmetric Rel$^n$ | These three cases are allowed in an Anti-symmetric rel$^n$ | | |

Q:. Let $A = \{1, 2, 3, \ldots, n\}$

How many "anti-symmetric relation" are possible on set A where $|A| = n$.

Number of Anti-symmetric Rel$^n$ = (Possible Combinations w.r.t. diagonal order pair) and (Possible Combinations w.r.t. non-diagonal order pair)

$$= 2^n * 3^{\left({}^nC_2\right)}$$

$$= \boxed{2^n * 3^{\frac{n(n-1)}{2}}}$$

Ans

Out of 'n' distinct elements, two distinct elements can be chosen in ${}^nC_2$ ways

Possible Choices of order pairs in an "anti-symmetric Relation" corresponding to each pair of distinct elements

# Topic : Asymmetric Relation

A relation 'R' on set A is called asymmetric relation if and only if,

$\left(\text{if } a \, R \, b, \text{ then } b \not R \, a\right), \forall a, b \in A$

i.e., $\left(\text{if } (a,b) \in R, \text{ then } (b,a) \notin R\right), \forall a, b \in A$

does not matter whether "$a = b$" or "$a \neq b$"

i.e., diagonal order pairs are definitely not allowed in an asymmetric relation

Slide

eg: Let $A = \{1, 2, 3\}$

① $R_1 = \{\ \}$ smallest asymmetric relation on set A

② $R_2 = \{(1,2), (3,3)\}$ Not an asymmetric relation

$(3,3) \rightarrow$ not allowed in asymmetric relⁿ.

③ $R_3 = \{(1,2), (2,3), (3,1), (3,2)\}$ Not asymmetric

$(2,3) \in R_3$ & $(3,2) \in R_3$

∴ Not Asymmetric

④ $R_4 = \{(1,2), (3,1), (3,2)\}$ it is asymmetric Relⁿ

Q: Let $A = \{1, 2, 3, \ldots, n\}$

How many "asymmetric relation" are possible on set A where $|A| = n$.

if $(a \neq b)$ then possible order pairs of a & b that are allowed in an asymmetric relation

$(a,b) \in R$ but $(b,a) \notin R$

$(a,b) \notin R$ and $(b,a) \in R$

$(a,b) \notin R$ and $(b,a) \notin R$

3 Combinations are allowed w.r.t two distinct elements

$(a,b) \in R$ and $(b,a) \in R$ not allowed

# Another approach w.r.t. non-diagonal order pairs

- In a non-diagonal order pair $(a,b)$, we know that $a \neq b$

  if $a=b$, then $(a,b)$ is a diagonal order pair.

- For a non-diagonal order pair we will require two distinct elements 'a' and 'b' s.t. $a \neq b$

- W.r.t. two distinct elements 'a' and 'b' possible combinations of order pairs in an arbitrary relation

| Case (i) | Case (ii) | Case (iii) | Case (iv) |
|---|---|---|---|
| both $(a,b)$, $(b,a)$ are present | $(a,b)$ is present but $(b,a)$ is not present | $(a,b)$ is not present but $(b,a)$ is present | Both $(a,b)$ and $(b,a)$ are not present |
| Not allowed in an asymmetric Relⁿ | These three cases are allowed in an asymmetric relⁿ | | |

Q:- Let $A = \{1, 2, 3, \ldots, n\}$

How many "asymmetric relation" are possible on set A where $|A| = n$.

Number of asymmetric $Rel^n$ = (Possible Combinations w.r.t. diagonal order pair) and (Possible Combinations w.r.t. non-diagonal order pair)

No diagonal order pair is allowed

$$= {}^{n}C_0 \quad * \quad 3^{\left({}^{n}C_2\right)}$$

out of 'n' distinct elements, two distinct elements can be chosen in ${}^{n}C_2$ ways

Possible Choices of order pairs in an "asymmetric Relation" corresponding to each pair of distinct elements

$$= \boxed{3^{\frac{n(n-1)}{2}}}$$

Ans

Let $A = \{1, 2, 3, 4, \ldots, n\}$

$$A \times A = \left\{ \begin{array}{cccccc} (1,1), & (1,2), & (1,3), & \ldots & (1,n) \\ (2,1), & (2,2), & (2,3), & \ldots & (2,n) \\ (3,1), & (3,2), & (3,3), & \ldots & (3,n) \\ \vdots & \vdots & \vdots & & \vdots \\ (n,1), & (n,2), & (n,3) & \ldots & (n,n) \end{array} \right\}$$

# 2 mins Summary

- **Topic** Irreflexive relation
- **Topic** Symmetric relation
- **Topic** Anti-symmetric relation
- **Topic** Asymmetric relation

Slide

# THANK - YOU

Slide